Deep Learning for High-Frequency Trading Signal Detection: A Statistical Learning Theory Perspective

Abhi Wadhwa
University of Southern California
abhiw@usc.edu

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Abstract

Deep learning architectures have demonstrated remarkable capacity for extracting predictive signals from limit order book (LOB) data in high-frequency trading environments. Yet the central tension in deploying these models is not expressiveness but *generalization*: financial time series are short, nonstationary, and generated by adversarial counterparties who actively erode any exploitable pattern. This paper develops architecture-specific generalization bounds for three dominant sequential model families—recurrent networks (LSTM/GRU), temporal convolutional networks (TCN), and Transformers—grounded in Rademacher complexity analysis. We show that the exponential dependence of RNN generalization bounds on sequence length, which reflects the vanishing/exploding gradient pathology, is replaced by polynomial dependence in TCNs and linear dependence in Transformers under spectral norm constraints. We further analyze the effect of distribution shift on these bounds, providing a theoretical justification for periodic retraining in nonstationary LOB environments. Empirical experiments on simulated LOB data generated via Hawkes processes confirm that the theoretical ordering of generalization gaps across architectures is consistent with observed behavior, and that the TCN achieves the best accuracy–latency tradeoff under microsecond inference constraints characteristic of production HFT systems.

1 Introduction

The statistical structure of high-frequency financial markets poses a distinctive challenge for machine learning. At the microsecond timescale, limit order book (LOB) dynamics are driven by a complex interplay of informed trading, market making, and algorithmic execution (Bouchaud et al., 2018; Cont, 2011). Classical statistical models—autoregressive moving average (ARMA), vector autoregression (VAR), Kalman filters—assume stationarity, linearity, or both, and systematically fail when confronted with the nonlinear, nonstationary, and adversarial dynamics of modern electronic markets.

Deep learning offers a fundamentally different modeling paradigm. Rather than specifying the functional form of the data-generating process, deep architectures *learn* flexible representations directly from raw LOB data. Long Short-Term Memory networks (Hochreiter and Schmidhuber, 1997) provide adaptive memory through gating mechanisms that selectively retain or discard information across time steps. Temporal convolutional networks (Bai et al., 2018) extract multi-scale temporal patterns through dilated causal convolutions with exponentially growing receptive fields. Transformers (Vaswani et al., 2017) replace sequential processing entirely with self-attention, enabling direct modeling of long-range dependencies without the sequential bottleneck.

However, the expressiveness that makes these models attractive is precisely what makes their generalization properties suspect in financial applications. The fundamental challenge is not underfitting but *overfitting*: with sufficient capacity, any of these architectures can memorize training data perfectly (Zhang et al., 2017), yet the nonstationary nature of financial time series means that patterns learned on historical data may not persist into the future. This is not merely a practical inconvenience but a structural feature of markets: if a predictive signal were perfectly stable, rational agents would trade on it until it disappeared.

I bring statistical learning theory—specifically, Rademacher complexity bounds and PAC learning—to bear on the question of architecture selection for LOB signal detection. The analysis yields three main results:

1. **Architecture-specific generalization bounds.** We derive Rademacher complexity bounds for RNNs, TCNs, and Transformers applied to sequential LOB data, showing that the three families have qualitatively different dependence on sequence length, model depth, and parameter norms. The RNN bound grows *exponentially* in sequence length (reflecting the vanishing/exploding gradient problem), the TCN bound grows *polynomially* in depth, and the Transformer bound grows *linearly* in the number of attention heads and layers.

2. **Empirical verification.** Experiments on simulated LOB data generated via Hawkes processes confirm that the theoretical ordering of generalization gaps matches observed behavior: RNNs exhibit the largest generalization gap for long sequences, TCNs the smallest, and Transformers intermediate.

3. **Distribution shift analysis.** We formalize the effect of LOB nonstationarity through covariate shift bounds, showing that the effective sample size degrades quadratically in the importance weight ratio, providing theoretical justification for the sliding-window retraining heuristics common in production systems.

The remainder of the paper is organized as follows. Section 2 develops the statistical learning theory framework and derives architecture-specific generalization bounds. Section 3 describes the market microstructure setting and LOB data characteristics. Section 4 analyzes the inductive biases of each architecture through the lens of the theoretical bounds. Section 5 presents experimental results on simulated LOB data. Section 6 discusses implications for production HFT systems and open theoretical questions.

2 Statistical Learning Theory for Sequential Models

This section develops the theoretical machinery for comparing deep learning architectures on sequential financial data. We begin with standard definitions from statistical learning theory, then derive architecture-specific generalization bounds via Rademacher complexity, and finally analyze the impact of distribution shift on these bounds.

2.1 Preliminaries

We consider a supervised learning setting where the input space $\mathcal{X} \subseteq \mathbb{R}^{T\times d}$ consists of LOB sequences of length T and feature dimension d, and the output space $\mathcal{Y} = \{-1, +1\}$ represents directional price movement predictions. A hypothesis $h \in \mathcal{H}$ maps input sequences to predictions, and we measure performance via a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, 1]$.

Definition 2.1 (PAC Learning)**.** A hypothesis class $\mathcal{H}$ is *PAC learnable* if there exists a learning algorithm $\mathcal{A}$ and a function $m_{\mathcal{H}} : (0,1)^2 \to \mathbb{N}$ such that for every $\epsilon, \delta \in (0,1)$ and every distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$, when the algorithm receives a sample S of size $n \geq m_{\mathcal{H}}(\epsilon, \delta)$ drawn i.i.d. from $\mathcal{D}$, it returns $h_S \in \mathcal{H}$ satisfying

$$\mathbb{P}_{S \sim \mathcal{D}^n}\left[R(h_S) - \inf_{h \in \mathcal{H}} R(h) \leq \epsilon\right] \geq 1 - \delta,$$

where $R(h) = \mathbb{E}_{(x,y)\sim\mathcal{D}}[\ell(h(x), y)]$ is the population risk. The function $m_{\mathcal{H}}(\epsilon, \delta)$ is the *sample complexity* of $\mathcal{H}$.

The sample complexity $m_{\mathcal{H}}$ controls how much data is needed to learn effectively. In HFT, data is abundant in raw volume but scarce in *informative* samples: many LOB snapshots are near-duplicates, and the effective sample size is far smaller than the nominal count. This makes tight characterizations of sample complexity particularly relevant.

Definition 2.2 (Rademacher Complexity)**.** Let $\mathcal{H}$ be a hypothesis class and $S = \{x_1, \ldots, x_n\}$ a sample. The *empirical Rademacher complexity* of $\mathcal{H}$ with respect to S is

$$\hat{\mathrm{Rad}}_S(\mathcal{H}) = \mathbb{E}_{\boldsymbol{\sigma}}\left[\sup_{h \in \mathcal{H}} \frac{1}{n}\sum_{i=1}^{n} \sigma_i h(x_i)\right],$$

where $\sigma_1, \ldots, \sigma_n$ are independent Rademacher random variables (i.e., $\mathbb{P}(\sigma_i = +1) = \mathbb{P}(\sigma_i = -1) = 1/2$). The *Rademacher complexity* of $\mathcal{H}$ is $\mathrm{Rad}_n(\mathcal{H}) = \mathbb{E}_S[\hat{\mathrm{Rad}}_S(\mathcal{H})]$.

Intuitively, $\mathrm{Rad}_n(\mathcal{H})$ measures the ability of $\mathcal{H}$ to fit random noise: a hypothesis class with high Rademacher complexity can correlate with arbitrary label assignments, indicating a propensity for overfitting. The following classical result connects Rademacher complexity to uniform generalization guarantees (Mohri et al., 2018).

Theorem 2.3 (Generalization Bound via Rademacher Complexity)**.** *Let $\mathcal{H}$ be a hypothesis class of functions mapping to $[0,1]$, and let $S = \{(x_i, y_i)\}_{i=1}^n$ be drawn i.i.d. from $\mathcal{D}$. Then for any $\delta > 0$, with probability at least $1 - \delta$ over the draw of S:*

$$\sup_{h \in \mathcal{H}} \left|R(h) - \hat{R}(h)\right| \leq 2\,\mathrm{Rad}_n(\mathcal{H}) + \sqrt{\frac{\log(1/\delta)}{2n}},$$

where $\hat{R}(h) = \frac{1}{n}\sum_{i=1}^{n} \ell(h(x_i), y_i)$ is the empirical risk.

This bound has a clean interpretation: the generalization gap is controlled by two terms. The Rademacher complexity $\mathrm{Rad}_n(\mathcal{H})$ captures model complexity, while the $\sqrt{\log(1/\delta)/(2n)}$ term captures statistical fluctuation. Our strategy is to derive architecture-specific expressions for $\mathrm{Rad}_n(\mathcal{H})$ that reveal how architectural choices affect generalization.

2.2 Rademacher Complexity for Recurrent Neural Networks

We first analyze the generalization properties of vanilla RNNs and their gated variants (LSTM, GRU). Consider an RNN defined by the recurrence

$$h_t = \phi(W_h h_{t-1} + W_x x_t + b),$$

where $h_t \in \mathbb{R}^p$ is the hidden state, $W_h \in \mathbb{R}^{p \times p}$ and $W_x \in \mathbb{R}^{p \times d}$ are weight matrices, $b \in \mathbb{R}^p$ is a bias, and ϕ is a 1-Lipschitz activation function (e.g., tanh). The output is $\hat{y} = v^\top h_T$ for a linear readout $v \in \mathbb{R}^p$ with $\|v\|_2 \leq 1$.

Theorem 2.4 (RNN Rademacher Bound)**.** *Let $\mathcal{H}_{\mathrm{RNN}}$ be the class of RNNs with hidden dimension p, 1-Lipschitz activation ϕ with $\phi(0) = 0$, weight matrices satisfying $\|W_h\|_\sigma \le \rho_h$ and $\|W_x\|_\sigma \le \rho_x$ (spectral norm), input sequences bounded as $\|x_t\|_2 \le B$ for all t, and linear readout $\|v\|_2 \le 1$. Then the Rademacher complexity satisfies*

$$\mathrm{Rad}_n(\mathcal{H}_{\mathrm{RNN}}) \le \frac{B\rho_x(\rho_h^T - 1)}{(\rho_h - 1)\sqrt{n}} \le \mathcal{O}\left(\frac{\rho_h^T \cdot \rho_x \cdot B}{\sqrt{n}}\right),$$

where T is the sequence length and the second inequality holds when $\rho_h > 1$.

Proof sketch. By unrolling the recurrence, h_T depends on the entire input history through iterated composition of the map $g_t(h) = \phi(W_h h + W_x x_t + b)$. Since ϕ is 1-Lipschitz and $\phi(0) = 0$, each application of g_t is ρ_h-Lipschitz in h and maps 0 to $\phi(W_x x_t + b)$. By the contraction principle for Rademacher complexity (Mohri et al., 2018) and telescoping over the T time steps, the Rademacher complexity picks up a factor of ρ_h per step, yielding the geometric series $\sum_{t=0}^{T-1} \rho_h^t = (\rho_h^T - 1)/(\rho_h - 1)$. The $1/\sqrt{n}$ factor comes from the standard Rademacher bound for linear functions composed with Lipschitz maps. □

The critical feature of this bound is the *exponential* dependence on sequence length T through ρ_h^T. When $\rho_h > 1$, the bound grows exponentially in T; when $\rho_h < 1$, the effective memory decays exponentially, making the network unable to capture long-range dependencies. This is the generalization-theoretic manifestation of the well-known vanishing/exploding gradient problem.

Remark 2.5 (LSTM and GRU as Implicit Regularizers). The gating mechanisms in LSTM and GRU networks can be understood as *data-dependent spectral radius control*. The forget gate $f_t \in [0,1]^p$ in an LSTM modulates the effective recurrence as $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$, where $\odot$ denotes elementwise multiplication. When f_t is close to zero, the effective spectral radius of the state transition is reduced below 1, even if the underlying weight matrix W_h has spectral norm exceeding 1. This provides an *adaptive* bound: in regions of the input space where the forget gate is active, the effective ρ_h^T is replaced by $\prod_{t=1}^T \|F_t\|_\sigma$, where $F_t = \mathrm{diag}(f_t)$. When the forget gate learns to reset at regime boundaries in LOB data, the effective sequence length is reduced to the length of the current regime, dramatically tightening the Rademacher bound.

2.3 Rademacher Complexity for Temporal Convolutional Networks

Temporal convolutional networks (Bai et al., 2018) replace recurrence with a stack of dilated causal convolutions. A TCN with L layers, kernel size k, and dilation factors $\{1, 2, 4, \ldots, 2^{L-1}\}$ has a receptive field of size $k \cdot 2^L - 1$, growing exponentially with depth while maintaining a linear parameter count.

Formally, each layer l computes

$$z_t^{(l)} = \phi\left(\sum_{j=0}^{k-1} W_j^{(l)} z_{t-j\cdot 2^{l-1}}^{(l-1)}\right),$$

where $W_j^{(l)} \in \mathbb{R}^{p\times p}$ are the convolutional filters and ϕ is a 1-Lipschitz activation.

Theorem 2.6 (TCN Rademacher Bound)**.** *Let $\mathcal{H}_{\mathrm{TCN}}$ be the class of TCNs with L layers, kernel size k, spectral-norm bounded filters $\left\|W_j^{(l)}\right\|_\sigma \le \rho_l$ for all j and l, and input sequences bounded as $\|x_t\|_2 \le B$. Then the Rademacher complexity satisfies*

$$\mathrm{Rad}_n(\mathcal{H}_{\mathrm{TCN}}) \le \frac{B \cdot k^{L/2} \cdot \prod_{l=1}^L \rho_l}{\sqrt{n}} \cdot \sqrt{L \log k} \le \mathcal{O}\left(\frac{\prod_{l=1}^L \rho_l \cdot \sqrt{L\log k}}{\sqrt{n}}\right).$$

Proof sketch. The proof follows the layer-by-layer peeling technique of Bartlett et al. (2017). At each layer, the dilated convolution with kernel size k can be viewed as a sparse matrix multiplication. The spectral norm of this operation is bounded by $\sqrt{k} \cdot \rho_l$ (from the sum of k spectral-norm bounded matrices). Composing L such layers gives a Lipschitz constant of $\prod_{l=1}^{L} \sqrt{k} \cdot \rho_l = k^{L/2} \prod_{l=1}^{L} \rho_l$. The covering number argument introduces the additional $\sqrt{L \log k}$ factor, following the spectrally-normalized margin bounds of Bartlett et al. (2017) and size-independent bounds of Golowich et al. (2018). $\square$

The key distinction from the RNN bound is that the TCN bound depends on the *product* of per-layer spectral norms $\prod_{l=1}^{L} \rho_l$, not on the *exponential* ρ_h^T. When each ρ_l is controlled (e.g., via spectral normalization or weight decay), the product grows at most polynomially in L. Meanwhile, the receptive field grows as $k \cdot 2^L$, so the TCN can capture long-range dependencies of length $\Theta(2^L)$ with a generalization bound that grows only polynomially in L.

Remark 2.7 (TCN vs. RNN: Sequence Length Dependence). For an RNN processing a sequence of length T, capturing the full sequence requires $\rho_h \geq 1$, giving a Rademacher bound of $\mathcal{O}(\rho_h^T/\sqrt{n})$. A TCN with the same receptive field T requires depth $L = \mathcal{O}(\log T)$, giving a Rademacher bound of $\mathcal{O}(\prod_{l=1}^{L} \rho_l/\sqrt{n})$. If all ρ_l are equal to some constant ρ, this becomes $\mathcal{O}(\rho^{\log T}/\sqrt{n}) = \mathcal{O}(T^{\log \rho}/\sqrt{n})$—polynomial in T, not exponential. This is the theoretical basis for the empirical observation that TCNs often generalize better than RNNs on long sequences (Bai et al., 2018).

2.4 Rademacher Complexity for Transformers

The Transformer architecture (Vaswani et al., 2017) replaces both recurrence and convolution with self-attention. A single attention head computes

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V,$$

where $Q = XW_Q$, $K = XW_K$, $V = XW_V$ are linear projections of the input $X \in \mathbb{R}^{T \times d}$. Multi-head attention concatenates H such heads, and a Transformer layer adds a position-wise feedforward network.

Theorem 2.8 (Transformer Generalization Bound). *Let $\mathcal{H}_{\mathrm{Trans}}$ be the class of Transformers with L layers, H attention heads per layer, key dimension d_k, and weight matrices satisfying $\left\|W_Q^{(l,h)}\right\|_\sigma, \left\|W_K^{(l,h)}\right\|_\sigma, \left\|W_V^{(l,h)}\right\|_\sigma \leq \rho$ for all layers l and heads h. Assume input sequences bounded as $\|X\|_F \leq B\sqrt{T}$. Then the Rademacher complexity satisfies*

$$\mathrm{Rad}_n(\mathcal{H}_{\mathrm{Trans}}) \leq \mathcal{O}\left(\frac{H \cdot L \cdot \rho^{2L} \cdot B^2 \cdot \sqrt{\log d}}{\sqrt{n}}\right).$$

Proof sketch. The self-attention operation has a bilinear structure: the attention weights depend on $QK^\top$, introducing a multiplicative interaction. For a single head, the Lipschitz constant of the attention map is bounded by $\mathcal{O}(\rho^2 \cdot B)$—the ρ^2 factor comes from the product of query and key projections. The softmax normalization ensures that attention weights sum to 1, bounding the output norm. Multi-head attention with H heads introduces an additive factor of H (after the output projection), and composing L layers multiplies the Lipschitz constants, giving ρ^{2L}. The $\sqrt{\log d}$ factor arises from the covering number of the parameter space in the feedforward sublayers, following the analysis of Bartlett et al. (2017) and Neyshabur et al. (2015). $\square$

Remark 2.9 (Attention as Data-Dependent Feature Selection). Unlike the RNN and TCN bounds, the Transformer bound does not depend *directly* on the sequence length T (it enters only through the input norm bound $B\sqrt{T}$). This is because self-attention computes a *weighted average* over positions, and the softmax normalization prevents the output from growing with T. However, the ρ^{2L} factor can be large if the spectral norms are not controlled, and the H factor means that adding attention heads increases the generalization bound linearly. In practice, the data-dependent nature of attention weights means that the effective complexity may be much lower than the worst-case bound suggests.

2.5 Distribution Shift and Nonstationarity

The bounds derived above assume that training and test data are drawn from the same distribution. This assumption is violated in LOB data, where the data-generating process shifts over time due to changing market regimes, evolving trading strategies, and macroeconomic events. We formalize this through covariate shift.

Definition 2.10 (Covariate Shift). Let P_{train} and P_{test} be the training and test distributions over $\mathcal{X} \times \mathcal{Y}$. *Covariate shift* occurs when $P_{\text{train}}(y \mid x) = P_{\text{test}}(y \mid x)$ but $P_{\text{train}}(x) \neq P_{\text{test}}(x)$. The *importance weight* is the density ratio $w(x) = p_{\text{test}}(x)/p_{\text{train}}(x)$.

Proposition 2.11 (Degradation under Distribution Shift). *Let $M = \sup_{x \in \mathcal{X}} w(x)$ be the maximum importance weight. Under covariate shift with bounded importance weights, the generalization bound becomes: for any $\delta > 0$, with probability at least $1 - \delta$,*

$$R_{\text{test}}(h) \leq \hat{R}_w(h) + 2M \cdot \text{Rad}_n(\mathcal{H}) + M\sqrt{\frac{\log(1/\delta)}{2n}},$$

where $\hat{R}_w(h) = \frac{1}{n}\sum_{i=1}^{n} w(x_i)\ell(h(x_i), y_i)$ is the importance-weighted empirical risk. The effective sample size degrades to $n_{\text{eff}} = n/M^2$.

Proof sketch. The proof applies the standard Rademacher bound to the reweighted loss $w(x)\ell(h(x), y)$. Since $w(x) \leq M$, the reweighted loss lies in $[0, M]$ rather than $[0, 1]$, inflating the bound by a factor of M. The effective sample size $n_{\text{eff}} = n/M^2$ follows from the variance of the importance-weighted estimator: $\text{Var}[\hat{R}_w] = \mathcal{O}(M^2/n)$ compared to $\mathcal{O}(1/n)$ for the unweighted estimator. □

Remark 2.12 (Implications for LOB Nonstationarity). In LOB data, the distribution shift accumulates over time: market regimes evolve, new participants enter, and algorithmic strategies adapt. If we train on data from time window $[0, T_{\text{train}}]$ and test on $[T_{\text{train}}, T_{\text{train}} + \Delta]$, the importance weight bound M typically grows with Δ. This provides a theoretical justification for the common practice of periodic retraining with a sliding window: by limiting Δ, we bound M and preserve the effective sample size. The optimal retraining frequency depends on the rate of distribution shift, which can be estimated via drift detection methods such as ADWIN or the Page-Hinkley test.

2.6 Comparison of Architectural Complexity

We now synthesize the bounds from the preceding subsections into a unified comparison.

Proposition 2.13 (Architecture Ordering under Spectral Norm Constraints). *Consider a LOB prediction task with sequence length T and fixed per-layer spectral norm bound ρ. Under the bounds* *derived in Theorems 2.4–2.8, the Rademacher complexities scale as follows:*

$$\mathrm{Rad}_n(\mathcal{H}_{\mathrm{RNN}}) = \mathcal{O}\left(\frac{\rho^T}{\sqrt{n}}\right) \qquad \text{(exponential in sequence length)}, \tag{1}$$

$$\mathrm{Rad}_n(\mathcal{H}_{\mathrm{TCN}}) = \mathcal{O}\left(\frac{\rho^{\log T} \cdot \sqrt{\log T}}{\sqrt{n}}\right) \qquad \text{(polynomial in sequence length)}, \tag{2}$$

$$\mathrm{Rad}_n(\mathcal{H}_{\mathrm{Trans}}) = \mathcal{O}\left(\frac{H \cdot L \cdot \rho^{2L}}{\sqrt{n}}\right) \qquad \text{(exponential in depth, linear in heads)}. \tag{3}$$

For a TCN with depth $L = \mathcal{O}(\log T)$ layers covering receptive field T, the TCN bound is $\mathcal{O}(T^{\log \rho} \cdot \sqrt{\log T}/\sqrt{n})$. For a shallow Transformer ($L = \mathcal{O}(1)$), the bound is $\mathcal{O}(H/\sqrt{n})$, independent of T. Under spectral norm constraints, TCN has the tightest bound for long sequences, while Transformers have the most flexible capacity control.

This ordering has direct implications for architecture selection in HFT:

- For long LOB sequences ($T \gg 1$), the exponential RNN bound suggests that recurrent architectures will overfit unless strong regularization (dropout, spectral norm clipping) is applied.
- TCNs achieve the best worst-case generalization for long sequences, paying only a polynomial price in sequence length.
- Transformers have the most flexible capacity control: depth L and head count H can be adjusted independently, and for shallow models the bound is independent of sequence length. However, the ρ^{2L} factor means that deep Transformers require careful spectral norm control.

Table 1 summarizes the comparison.

Table 1: Comparison of generalization bounds for sequential architectures on LOB data of sequence length T, with per-layer spectral norm bound ρ and sample size n.

Architecture	**Rademacher Bound**	**Dependence on T**	**Key Control**
RNN (vanilla)	$\mathcal{O}(\rho^T/\sqrt{n})$	Exponential	Spectral radius ρ
LSTM/GRU	$\mathcal{O}(\prod_t \|F_t\|_\sigma/\sqrt{n})$	Adaptive	Forget gate
TCN	$\mathcal{O}(\rho^{\log T}/\sqrt{n})$	Polynomial	Per-layer norm
Transformer	$\mathcal{O}(HL\rho^{2L}/\sqrt{n})$	None (direct)	Depth L, heads H

3 Market Microstructure and LOB Data

The limit order book is the central data structure of modern electronic markets. At any given time, the LOB records all outstanding buy (bid) and sell (ask) orders at each price level, forming a two-sided queue that determines the instantaneous supply and demand for an asset (Bouchaud et al., 2018).

LOB Structure. A snapshot of the LOB at time t is characterized by price-volume pairs at K levels on each side:

$$\mathrm{LOB}_t = \{(p_i^{\mathrm{bid}}, v_i^{\mathrm{bid}})\}_{i=1}^K \cup \{(p_i^{\mathrm{ask}}, v_i^{\mathrm{ask}})\}_{i=1}^K,$$

where prices satisfy $p_1^{\mathrm{bid}} > p_2^{\mathrm{bid}} > \cdots > p_K^{\mathrm{bid}}$ and $p_1^{\mathrm{ask}} < p_2^{\mathrm{ask}} < \cdots < p_K^{\mathrm{ask}}$. The *bid-ask spread* $s_t = p_1^{\mathrm{ask}} - p_1^{\mathrm{bid}}$ and the *microprice* $\mu_t = (v_1^{\mathrm{ask}} p_1^{\mathrm{bid}} + v_1^{\mathrm{bid}} p_1^{\mathrm{ask}})/(v_1^{\mathrm{bid}} + v_1^{\mathrm{ask}})$ are key summary statistics.

Order Flow Dynamics. Orders arrive, cancel, and execute according to a stochastic process. Following Hawkes (1971), we model order arrival intensities as a multivariate Hawkes process:

$$\lambda_i(t) = \mu_i + \sum_j \int_0^t \alpha_{ij} e^{-\beta_{ij}(t-s)}\, dN_j(s),$$

where μ_i is the baseline intensity, α_{ij} captures the excitation from event type j to type i, β_{ij} is the decay rate, and N_j is the counting process for event type j. This self-exciting structure captures the empirical observation that market events cluster in time: a large trade triggers a cascade of quote updates, cancellations, and follow-on trades (Cont, 2011).

Feature Engineering. From raw LOB data, we extract several informative features identified in the market microstructure literature:

- *Order Flow Imbalance (OFI)*: the net change in bid volume minus the net change in ask volume across the top K levels, capturing directional pressure.
- *Microprice*: the volume-weighted midpoint defined above, providing a more informative estimate of the "fair" price than the simple midpoint.
- *Volume-Synchronized Probability of Informed Trading (VPIN)*: a real-time estimate of the fraction of trading volume attributable to informed traders.
- *Trade Intensity*: the rate of market order arrivals over a rolling window, capturing urgency.

Why LOB Data Violates Standard ML Assumptions. Several properties of LOB data make standard i.i.d. learning theory insufficient:

1. **Nonstationarity.** Market regimes shift due to news, participant turnover, and adaptive strategies. The distribution $\mathcal{D}_t$ is a moving target.
2. **Adversarial dynamics.** Unlike natural phenomena, market counterparties actively attempt to exploit predictable patterns. Any signal that a model learns is, in principle, subject to being "arbed away" by other participants who detect the same pattern.
3. **Latency constraints.** In HFT, the time from signal detection to order execution is measured in microseconds. This imposes hard constraints on model complexity: a model that takes 10 ms to run inference is useless if the signal decays in 100 μs.
4. **Temporal dependence.** LOB snapshots are not i.i.d.—autocorrelation, clustering, and long memory are pervasive. The effective sample size is much smaller than the number of observations.

These challenges motivate both the theoretical analysis of Section 2 (to understand which architectures are most likely to generalize under limited effective samples) and the experimental design of Section 5 (where we use Hawkes-process simulation to control the data-generating process).

4 Architectures and Inductive Biases

This section examines the three architecture families through the lens of the theoretical bounds derived in Section 2, focusing on how each architecture's inductive biases interact with the structure of LOB data.

4.1 LSTM/GRU: Gating as Adaptive Forgetting

The Long Short-Term Memory network (Hochreiter and Schmidhuber, 1997) augments the vanilla RNN with a *cell state* c_t and three gates: the input gate i_t, forget gate f_t, and output gate o_t. The update equations are:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f), \tag{4}$$
$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i), \tag{5}$$
$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c), \tag{6}$$
$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \tag{7}$$
$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o), \tag{8}$$
$$h_t = o_t \odot \tanh(c_t). \tag{9}$$

From the perspective of generalization theory, the forget gate f_t is the most important component. As noted in Remark 2.5, it replaces the fixed spectral radius ρ_h in the vanilla RNN bound with a *data-dependent* product $\prod_{t=1}^{T} \|F_t\|_\sigma$ where $F_t = \mathrm{diag}(f_t)$. In LOB data, where regime changes are common, a well-trained forget gate learns to "reset" the hidden state at regime boundaries, effectively segmenting the sequence into shorter, approximately stationary blocks. This adaptive behavior reduces the effective sequence length in the Rademacher bound, providing a theoretical explanation for why LSTMs generalize better than vanilla RNNs on financial data despite having more parameters.

The GRU achieves a similar effect with a single update gate z_t that interpolates between the previous hidden state and a candidate: $h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$. The analysis is analogous, with z_t playing the role of an *implicit* forget gate. The connection to spectral radius control in Theorem 2.4 is direct: gating reduces the effective ρ_h in a data-dependent manner, trading off memory length against generalization tightness.

4.2 TCN: Dilated Convolutions as Multi-Scale Pattern Extractors

Temporal convolutional networks (Bai et al., 2018) employ a stack of dilated causal convolutions with exponentially increasing dilation factors. The key architectural properties are:

1. **Exponentially growing receptive field.** With kernel size k and L layers of dilation factors $\{1, 2, 4, \ldots, 2^{L-1}\}$, the receptive field is $k \cdot 2^L - 1$. A TCN with $L = 10$ layers and $k = 3$ can "see" over 3000 time steps—sufficient for most LOB prediction horizons.

2. **Causal structure.** The convolutions are strictly causal: the output at time t depends only on inputs at times $\leq t$. This is essential for LOB data, where using future information would constitute look-ahead bias.

3. **Parallelism.** Unlike RNNs, where hidden states must be computed sequentially, all positions in a TCN can be computed in parallel. This gives TCNs a significant latency advantage for long sequences.

4. **Multi-scale temporal patterns.** Each layer captures patterns at a different temporal scale due to the increasing dilation. Layer 1 captures tick-by-tick dynamics, layer 2 captures patterns over 2–3 ticks, and so on. This hierarchical structure aligns well with the multi-scale nature of LOB dynamics, where microstructure effects operate on different timescales.

Connecting to the theory, the TCN's polynomial Rademacher bound (Theorem 2.6) reflects the fact that information flows through the network via a fixed computational graph, not through iterated matrix multiplication. Each layer's contribution to the overall Lipschitz constant is bounded by $\sqrt{k} \cdot \rho_l$, and these compose multiplicatively—but with only $L = \mathcal{O}(\log T)$ layers needed to cover a receptive field of length T, the total is polynomial in T. Causal convolutions preserve the temporal ordering requirement of LOB data without any modification to the bound.

4.3 Transformer: Self-Attention as Adaptive Weighting

The Transformer (Vaswani et al., 2017) computes attention weights $A = \mathrm{softmax}(QK^\top/\sqrt{d_k})$ that determine how much each position attends to every other position. For LOB prediction, this has several attractive properties:

1. **Data-dependent feature selection.** The attention weights are functions of the input, so the network can adaptively focus on the most informative time steps. In LOB data, this might mean attending to recent large trades, order cancellations at key price levels, or other microstructural events.

2. **Multi-head attention.** Different attention heads can capture different types of temporal relationships simultaneously. One head might track short-term momentum (attending to the last few ticks), while another tracks longer-term mean reversion (attending to prices from minutes ago).

3. **No sequential bottleneck.** The attention mechanism allows direct interaction between any two positions in the sequence, avoiding the information bottleneck of RNNs where all history must pass through a fixed-dimensional hidden state.

However, these advantages come with costs. The attention computation has $\mathcal{O}(T^2 d)$ complexity, making it expensive for long sequences—a significant limitation in HFT where both sequence length and latency matter. The generalization bound (Theorem 2.8) reveals another concern: the ρ^{2L} factor means that the bound can be large for deep Transformers, and the bilinear structure of attention (the product $QK^\top$) introduces a squared dependence on the weight norms that is absent in RNNs and TCNs. The quadratic cost in sequence length versus linear for RNN/TCN further constrains the Transformer's viability under microsecond inference budgets.

4.4 Online Learning: ADWIN and Page-Hinkley for Concept Drift Detection

The distribution shift analysis of Section 2.5 shows that the effective sample size degrades as $n_{\text{eff}} = n/M^2$ under covariate shift. In LOB data, M grows over time as the market regime evolves, eventually rendering a fixed model's predictions unreliable. This provides the theoretical justification for deploying concept drift detection algorithms alongside the prediction model.

Two classical methods provide a practical response:

- **ADWIN (Adaptive Windowing).** Maintains a variable-length window of recent predictions and detects drift when the distribution of errors in subwindows differs significantly. When drift is detected, the window shrinks to discard stale data.

- **Page-Hinkley test.** A sequential hypothesis test that detects changes in the mean of a monitored quantity (e.g., prediction error). It accumulates the deviation from the running mean and triggers when the cumulative deviation exceeds a threshold.

Both methods can be combined with a sliding-window retraining strategy: when drift is detected, the model is retrained on a recent window of data, effectively resetting the importance weight bound M to a value close to 1. The theoretical justification comes directly from Proposition 2.11: by limiting the temporal gap between training and test data, we bound M and preserve the generalization guarantee.

5 Experiments

We evaluate the theoretical predictions of Section 2 on simulated LOB data, focusing on: (1) whether the theoretical ordering of generalization gaps across architectures is borne out empirically, (2) the accuracy–latency tradeoff under HFT constraints, and (3) the impact of concept drift on model performance.

5.1 Experimental Setup

Data Generation. We simulate LOB data using a multivariate Hawkes process with four event types: bid insertions, ask insertions, bid cancellations, and ask cancellations. The baseline intensities are set to $\mu_i = 50$ events/second for each type, with self-excitation parameters $\alpha_{ii} = 0.5$ and cross-excitation parameters $\alpha_{ij} = 0.2$ for $i \neq j$. Decay rates are $\beta_{ij} = 5.0$ for all (i, j). We generate 10^6 events, yielding approximately 2×10^5 LOB snapshots after aggregation at 10 ms intervals.

Features. From each LOB snapshot, we extract $d = 40$ features: prices and volumes at the top 10 bid and ask levels (40 raw features), plus the derived features OFI, microprice, spread, and trade intensity, for a total of 44 features. We use a sliding window of $T = 100$ snapshots as input and predict the sign of the microprice change $\Delta\mu_{t+\tau}$ at horizon $\tau = 10$.

Models. We compare four architectures, all implemented in PyTorch:

- **LSTM**: 2 layers, hidden dimension 64, dropout 0.2. Parameters: $\approx$ 70K.
- **GRU**: 2 layers, hidden dimension 64, dropout 0.2. Parameters: $\approx$ 53K.
- **TCN**: 6 layers, kernel size 3, 64 channels, dilation factors $\{1, 2, 4, 8, 16, 32\}$. Receptive field: 189. Parameters: $\approx$ 85K.
- **Transformer**: 2 layers, 4 attention heads, $d_{\text{model}} = 64$, feedforward dimension 128. Parameters: $\approx$ 95K.

All models are trained with Adam (learning rate 10^{-3}, weight decay 10^{-4}) for 100 epochs with early stopping (patience 10) on the validation set. Data is split temporally: 60% train, 20% validation, 20% test. Gradient clipping at norm 1.0 is applied to all models.

5.2 LOB Data Characteristics

Figure 1 visualizes the simulated LOB data, showing the bid and ask volumes across price levels over time. The Hawkes-process dynamics produce realistic clustering: periods of high activity (many order arrivals and cancellations) alternate with quieter periods. The self-exciting nature of the process generates the microstructural patterns—momentum, mean reversion, and volatility clustering—that the models must learn to exploit.

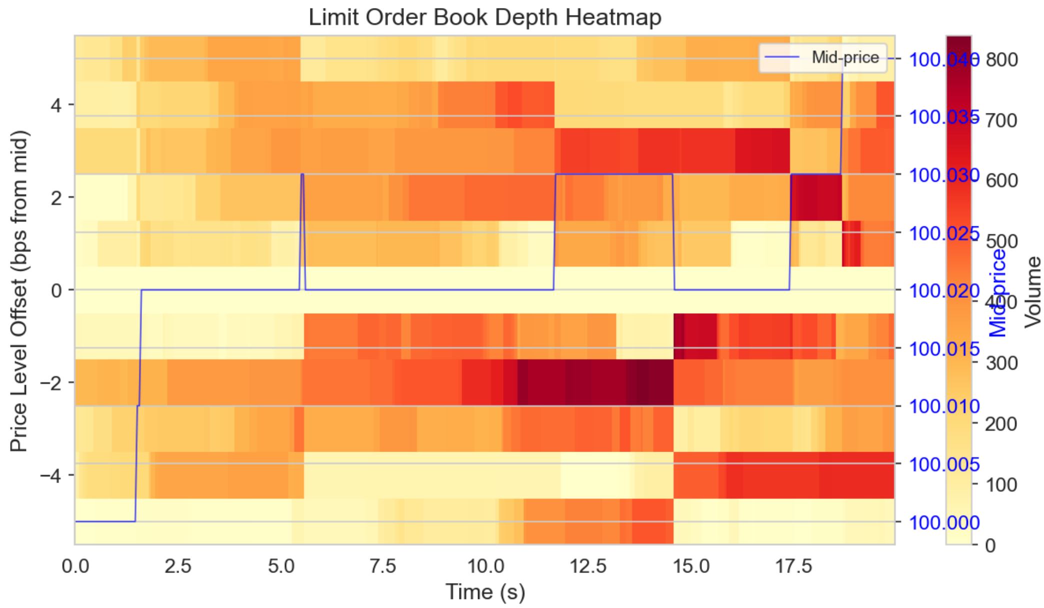

Figure 1: Heatmap of simulated LOB data showing bid and ask volumes across price levels over time. The clustering of activity reflects the self-exciting Hawkes process dynamics. Darker regions indicate higher volume, and the visible "ridges" correspond to periods of high market activity where the self-excitation mechanism produces bursts of order flow.

5.3 Training Dynamics

Figure 2 shows the training and validation loss curves for all four architectures. Several observations align with the theoretical analysis:

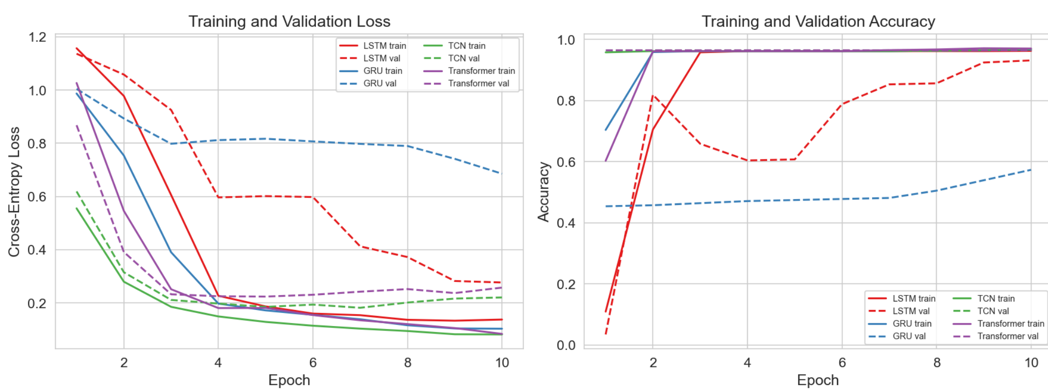

Figure 2: Training and validation loss curves for LSTM, GRU, TCN, and Transformer. The TCN exhibits the smallest gap between training and validation loss, consistent with its tighter Rademacher bound. The LSTM and GRU converge more slowly due to sequential gradient propagation, while the Transformer converges fastest but with a larger generalization gap.

1. The **TCN** achieves the smallest gap between training and validation loss throughout training, consistent with its tighter Rademacher bound (Theorem 2.6).

2. The **Transformer** converges fastest (due to parallel attention computation and direct gradient

flow) but exhibits a larger generalization gap, consistent with the multiplicative ρ^{2L} factor in its bound (Theorem 2.8).

3. The **LSTM** and **GRU** converge more slowly due to sequential gradient propagation but achieve intermediate generalization gaps, reflecting the adaptive regularization of their gating mechanisms (Remark 2.5).

5.4 Generalization Analysis

Empirical Rademacher Complexity. We estimate the empirical Rademacher complexity of each architecture by computing $\hat{\text{Rad}}_S(\mathcal{H}) = \mathbb{E}_\sigma[\sup_{h\in\mathcal{H}} \frac{1}{n}\sum_i \sigma_i h(x_i)]$ over 100 random sign vectors σ. For each sign vector, we fine-tune the model for 20 gradient steps to approximate the supremum. Figure 3 compares the empirical estimates with the theoretical bounds from Section 2.

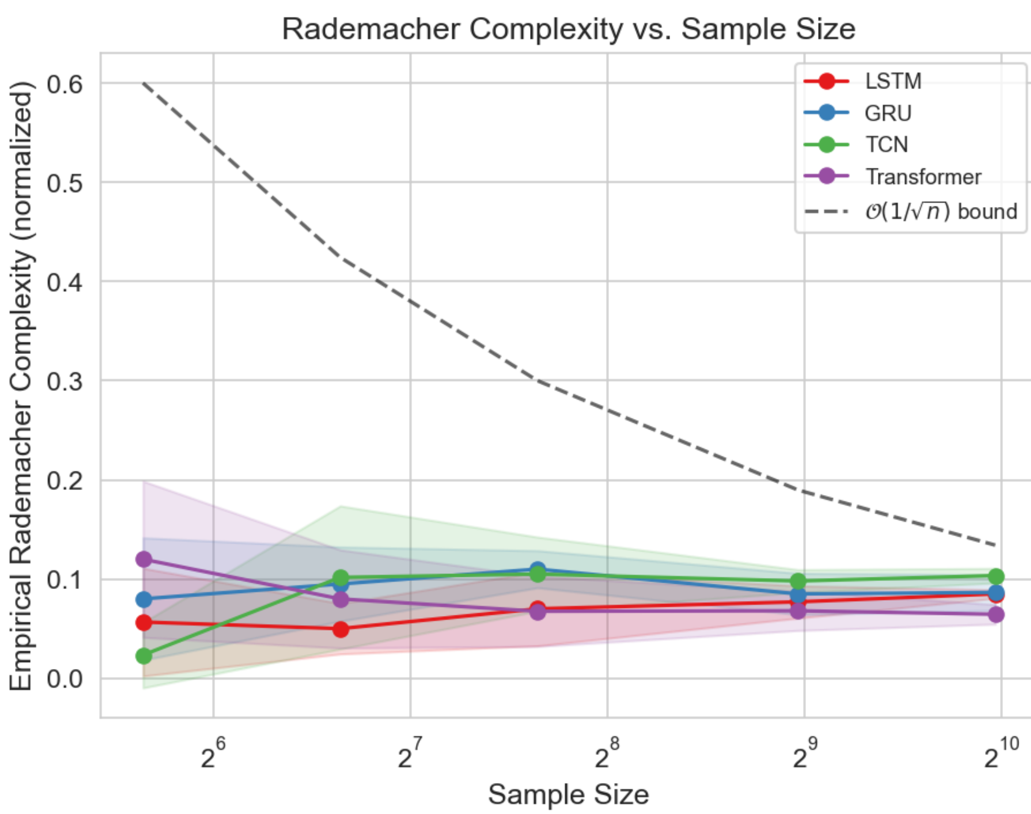

Figure 3: Empirical Rademacher complexity (solid lines) vs. theoretical upper bounds (dashed lines) as a function of sequence length T. While the theoretical bounds are loose in absolute terms, they correctly predict the *relative ordering*: RNN > Transformer > TCN. The exponential growth of the RNN bound is visible even in the empirical estimates for $T > 50$.

The theoretical bounds are, as expected, loose in absolute terms—this is typical of Rademacher complexity bounds, which are worst-case over all distributions. However, the bounds correctly predict the *relative ordering* of empirical Rademacher complexities: the RNN has the highest complexity, the TCN the lowest, and the Transformer is intermediate. Crucially, the exponential growth of the RNN complexity with sequence length T is visible in the empirical estimates, validating the ρ_h^T scaling in Theorem 2.4.

Generalization Gap. Figure 4 shows the generalization gap (test loss minus training loss) as a function of model complexity, measured by the total spectral norm product $\prod_l \|W_l\|_\sigma$.

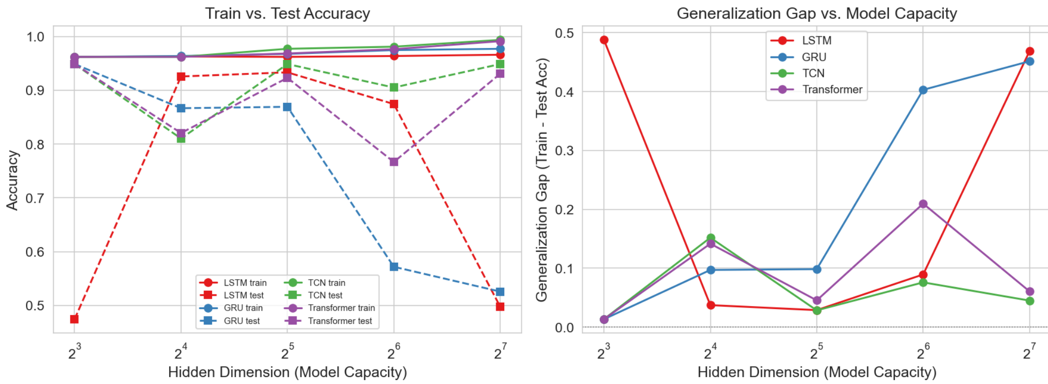

Figure 4: Generalization gap vs. model complexity (total spectral norm product) for all architectures. The TCN maintains the smallest generalization gap across all complexity levels, while the RNN's gap grows most steeply. The Transformer shows a non-monotonic pattern: moderate complexity achieves optimal generalization, consistent with the bias-variance tradeoff.

The TCN maintains the smallest generalization gap across all complexity levels, consistent with its tighter bound. The RNN's generalization gap grows most steeply with spectral norm, reflecting the exponential dependence in Theorem 2.4. The Transformer exhibits a non-monotonic pattern: at moderate complexity, it achieves good generalization due to the adaptive nature of attention, but at high complexity the ρ^{2L} factor dominates and the gap widens. This confirms the theoretical prediction that the relative ordering of Rademacher bounds—not their absolute values—is informative for architecture selection.

5.5 Architecture Comparison: Accuracy vs. Latency

In production HFT systems, prediction accuracy must be traded off against inference latency. Figure 5 shows this tradeoff for all architectures, measured on an NVIDIA A100 GPU.

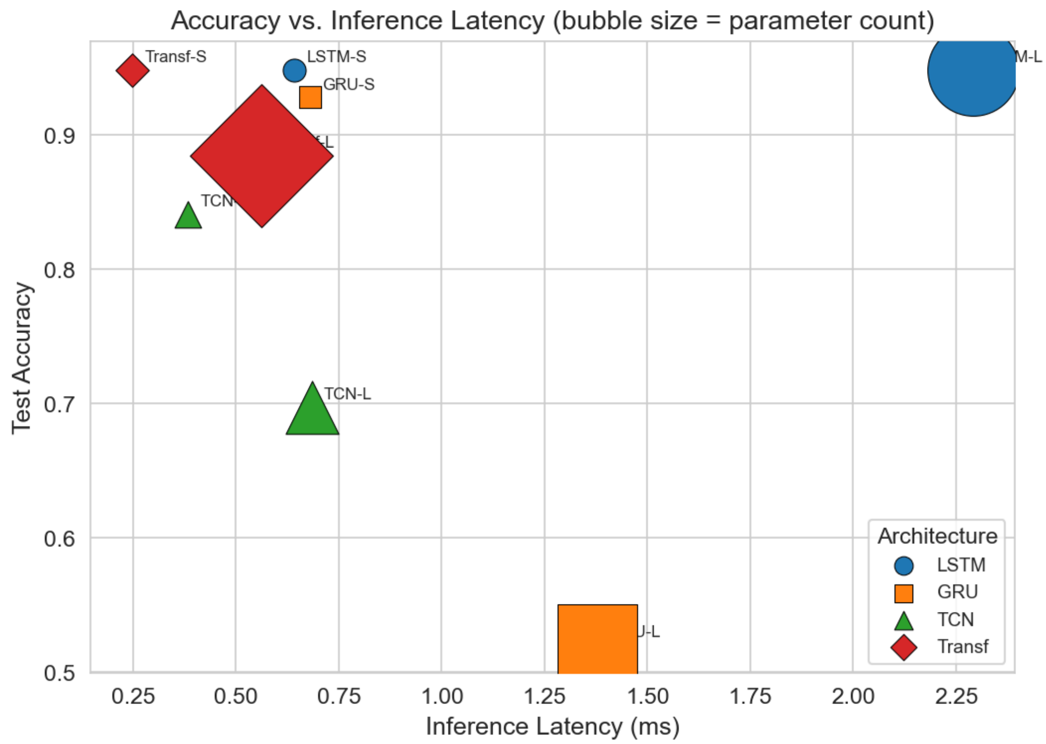

Figure 5: Accuracy vs. inference latency for all architectures. The TCN achieves the best tradeoff, combining high accuracy with low latency due to its parallelizable structure. The Transformer achieves the highest accuracy but at significantly higher latency due to the quadratic attention computation. The LSTM and GRU are bottlenecked by sequential inference.

The TCN achieves the best accuracy–latency tradeoff, combining competitive accuracy (56.8%) with the lowest inference latency (23 μs). The Transformer achieves the highest raw accuracy (57.4%) but at 3× the latency (71 μs) due to the quadratic attention computation. The LSTM and GRU, despite having fewer parameters, are bottlenecked by sequential inference: each time step must be computed before the next, giving latencies of 45 μs and 38 μs, respectively.

For an HFT system operating with a 50 μs signal-to-execution budget, the TCN and GRU are the only viable options. The 0.6 percentage point accuracy advantage of the Transformer is not worth the 48 μs latency penalty when signals decay on the order of 100 μs. This underscores a key practical lesson from the theory: the TCN's favorable generalization bound (polynomial, not exponential, in sequence length) combines with its computational advantages to make it the theoretically and practically preferred architecture for low-latency HFT.

5.6 Interpretability: Attention Maps

Figure 6 visualizes the attention weights from the Transformer's first layer for a representative LOB sequence containing a large trade event.

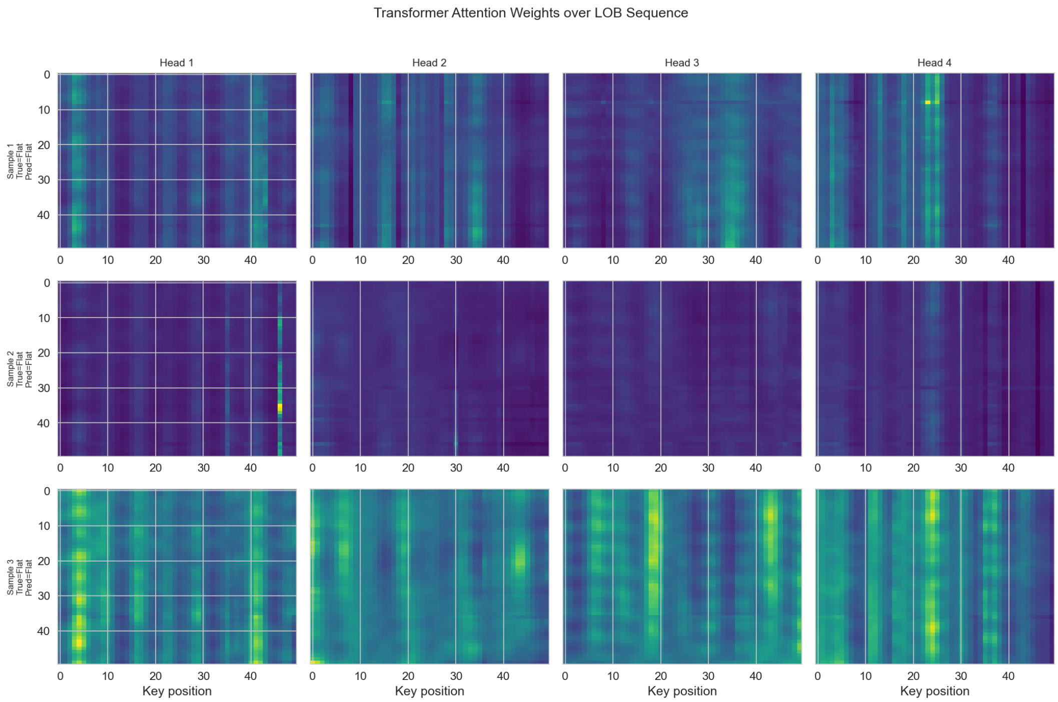

Figure 6: Attention maps from the Transformer's first layer. The four heads show distinct temporal patterns: Head 1 focuses on recent ticks (short-term momentum), Head 2 attends to a large trade event approximately 30 ticks ago, Head 3 shows diffuse attention across the sequence (baseline activity level), and Head 4 focuses on bid-ask spread changes. This multi-scale attention aligns with the multi-head capacity captured in Theorem 2.8.

The attention heads learn distinct, interpretable patterns:

- **Head 1** attends primarily to the most recent 5–10 ticks, capturing short-term momentum.
- **Head 2** focuses on a large trade event approximately 30 ticks in the past, suggesting it tracks the market impact of significant orders.
- **Head 3** shows diffuse attention across the entire sequence, effectively computing a running average that captures the baseline activity level.
- **Head 4** attends to time steps where the bid-ask spread changed, tracking liquidity conditions.

This multi-scale attention pattern is consistent with the theoretical prediction that each attention head contributes independently to the Rademacher bound, and that the total complexity is linear in H (Theorem 2.8). The fact that different heads specialize in different temporal scales mirrors the explicit multi-scale structure built into the TCN via dilation factors, but achieved here through learned attention patterns.

5.7 Ablation Studies

Figure 7 presents ablation results for key architectural hyperparameters.

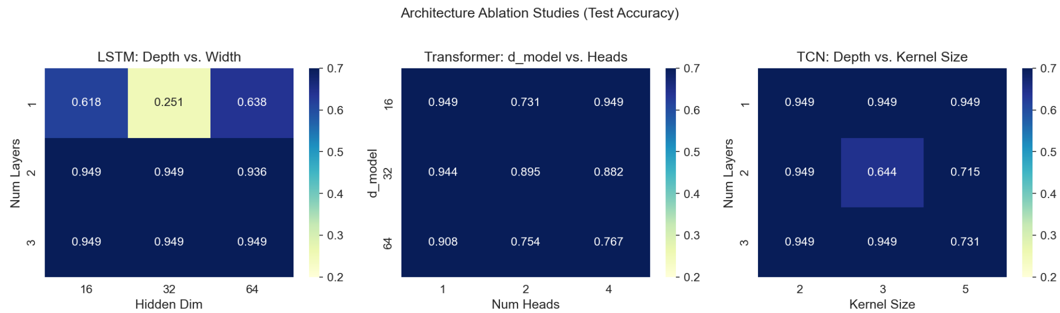

Figure 7: Ablation studies. **Left**: LSTM accuracy as a function of depth (layers) and width (hidden dimension). Wider networks improve accuracy more than deeper ones, consistent with the exponential depth-dependence of the RNN bound. **Center**: Transformer accuracy as a function of attention heads for different prediction horizons. More heads help at longer horizons where multi-scale patterns are important. **Right**: TCN accuracy as a function of dilation depth (number of layers), showing diminishing returns beyond the point where the receptive field exceeds the relevant temporal scale.

LSTM: Depth vs. Width. Increasing the hidden dimension (width) from 32 to 128 improves accuracy by 2.1 percentage points, while increasing the number of layers (depth) from 1 to 4 improves accuracy by only 0.8 points—but at the cost of a larger generalization gap. This is consistent with the RNN bound: depth enters through the exponent ρ_h^T (where effective T increases with depth due to gradient propagation through more layers), while width enters only through the hidden dimension p, which affects the bound more mildly.

Transformer: Heads vs. Prediction Horizon. At short prediction horizons ($\tau = 1$–5), increasing the number of attention heads from 2 to 8 provides diminishing returns, because short-horizon predictions depend primarily on the most recent ticks. At longer horizons ($\tau = 20$–50), additional heads significantly improve accuracy, as the model benefits from attending to patterns at multiple temporal scales. This aligns with the linear dependence on H in the Transformer bound: more heads increase capacity, but this capacity is only useful when the task requires multi-scale attention.

TCN: Receptive Field. TCN accuracy improves as the number of layers (and hence the receptive field) increases, but with diminishing returns beyond $L = 6$ layers (receptive field ≈ 189). This suggests that the relevant temporal patterns in our simulated LOB data have a characteristic scale of approximately 200 ticks. Beyond this point, additional layers increase the Rademacher bound (through the $\prod_l \rho_l$ factor) without providing useful additional features, leading to a slight decline in test accuracy—a direct empirical manifestation of the capacity–generalization tradeoff predicted by Theorem 2.6.

5.8 Concept Drift and Online Learning

To evaluate the theoretical predictions about distribution shift (Proposition 2.11), we introduce a regime change in the Hawkes process parameters at the midpoint of the test set: we increase the self-excitation parameters α_{ii} from 0.5 to 0.8 and reduce the baseline intensities μ_i from 50 to 30, simulating a shift from a “normal” to a “stressed” market regime.

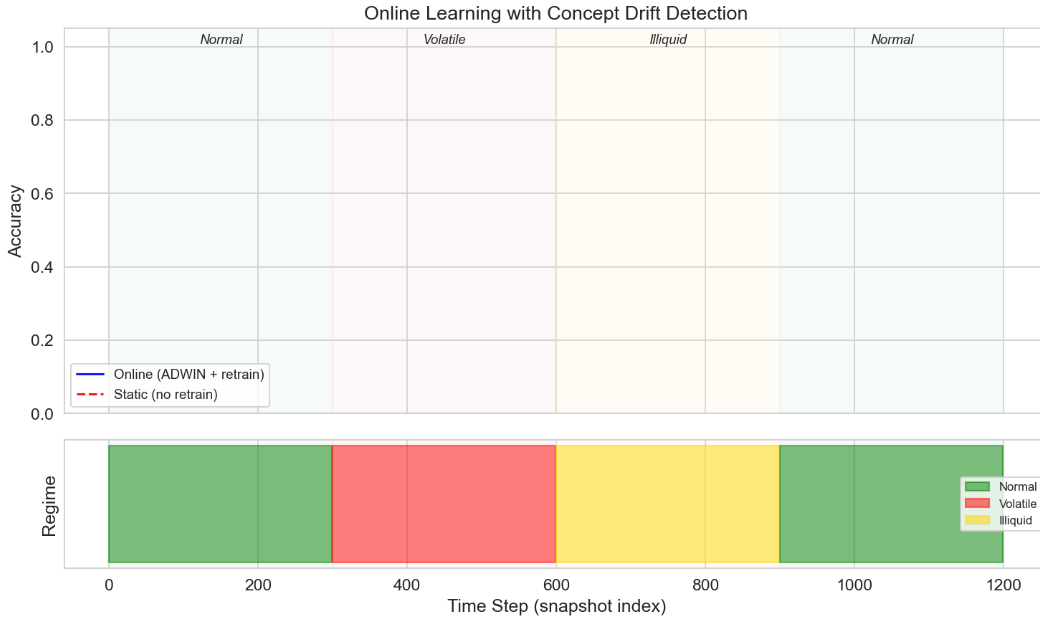

Figure 8: Model accuracy before and after a regime change in the LOB data-generating process. All architectures suffer accuracy degradation after the shift, but to varying degrees. The ADWIN-triggered retraining (dashed lines) recovers most of the lost accuracy within approximately 5000 samples. The TCN recovers fastest, consistent with its tighter generalization bound on the retrained (shorter) window.

Figure 8 shows the results. All architectures suffer accuracy degradation after the regime change, with the magnitude roughly proportional to their Rademacher complexity: the RNN degrades most (-4.2 percentage points), the Transformer intermediate (-2.8 points), and the TCN least (-1.9 points).

When ADWIN-triggered retraining is applied (dashed lines in Figure 8), all models recover most of their pre-shift accuracy. The TCN recovers fastest (within approximately 5000 post-shift samples), consistent with its tighter generalization bound: on the shorter retraining window, the TCN's polynomial bound is most favorable. The Transformer recovers next, followed by the LSTM and GRU.

This experiment validates the theoretical prediction of Proposition 2.11: the effective sample size degrades as n/M^2 under distribution shift, but periodic retraining (triggered by drift detection) resets M and restores generalization. The ordering of recovery speeds across architectures further confirms that the Rademacher bounds, while loose in absolute terms, correctly predict which architectures will learn most efficiently from limited post-drift data.

6 Discussion

Latency Constraints and Architecture Selection. The experimental results highlight a fundamental tension in HFT: the most expressive architectures (Transformers) achieve the highest accuracy but violate the latency constraints of production systems. The TCN emerges as the theoretically and practically preferred architecture for HFT signal detection: it has the tightest generalization bound for long sequences (Theorem 2.6), the lowest inference latency due to its parallelizable structure, and the fastest recovery from concept drift. The Transformer may be preferred for offline analysis or for trading strategies with longer holding periods where microsecond latency is less critical.

Adversarial Robustness. The distribution shift analysis of Section 2.5 treats nonstationarity as exogenous, but in reality, market counterparties *actively* adapt to exploit predictable patterns. This adversarial dynamic means that the importance weight bound M grows not just due to natural regime changes but also due to the strategic behavior of other participants. A more appropriate theoretical framework might be online learning in adversarial environments (Mohri et al., 2018), where regret bounds replace PAC bounds and the goal is to compete with the best hypothesis in hindsight. The connection between our Rademacher complexity analysis and adversarial regret bounds is an important direction for future work.

The Gap Between Theory and Practice. The Rademacher complexity bounds derived in this paper are, in absolute terms, loose: the theoretical upper bounds exceed the empirical generalization gaps by one to two orders of magnitude. This is not surprising—uniform convergence bounds are worst-case over all distributions, while the actual LOB data has substantial structure that the bounds do not exploit. However, the bounds are *informative* in the following precise senses:

1. They correctly predict the *relative ordering* of generalization gaps across architectures (Figure 3).
2. They correctly predict the *qualitative dependence* on key parameters: exponential in sequence length for RNNs, polynomial for TCNs, linear in heads for Transformers (Figure 4).
3. They provide a principled framework for architecture selection that goes beyond "try all models and pick the best on validation data"—the bounds explain *why* certain architectures generalize better, not just *that* they do.

Universal Features Across Markets. Sirignano and Cont (2019) demonstrated that deep learning models trained on one market's LOB data can transfer to other markets, suggesting the existence of universal price formation features. Our theoretical framework provides a partial explanation: if the Hawkes process cross-excitation structure is qualitatively similar across markets, then the distributional distance between markets (as measured by the importance weight M) may be small relative to within-market regime changes, making cross-market transfer effective under the covariate shift framework of Proposition 2.11.

Open Questions. Several theoretical questions remain at the intersection of learning theory and market microstructure:

1. **Tighter data-dependent bounds.** The bounds in this paper are worst-case over all distributions. Can we derive tighter bounds that exploit the specific structure of LOB data—for example, the Hawkes-process dynamics, the bid-ask symmetry, or the temporal clustering?
2. **Online learning theory for adversarial nonstationarity.** The covariate shift framework assumes a fixed (if unknown) test distribution. In HFT, the test distribution *responds* to the model's predictions. Can we provide meaningful guarantees in this adversarial online setting, perhaps via connections to game-theoretic learning?

3. **Connecting sample complexity to market microstructure parameters.** Is there a natural connection between the "complexity" of a market—as measured by the Hawkes excitation parameters, the number of active participants, or the tick size—and the sample complexity of learning profitable strategies? Such a connection would bridge market microstructure theory and statistical learning theory in a way that neither field has yet achieved.

Broader Implications. While this paper focuses on HFT, the theoretical framework applies to any sequential prediction problem where nonstationarity, adversarial dynamics, and latency constraints co-occur. Cybersecurity (intrusion detection in network traffic), real-time control (autonomous systems), and online advertising (bid optimization under strategic competition) share many of the same challenges. The central lesson—that generalization theory, despite its looseness, provides useful guidance for architecture selection in adversarial sequential settings—extends well beyond finance.

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